

Assignment No. 7

Subject: Database Engineering Lab (7CS271)

Class: SY CSE, AY 2025–26, Sem-II

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Batch - S5

Functional dependencies and attribute closure

Q. Analyze functional dependencies in relational schemas and compute attribute closures to determine candidate keys and dependency implications.

Consider the relation: R (A, B, C, D, E) Functional dependencies:

$A \rightarrow B$ $BC \rightarrow D$ $D \rightarrow E$ $E \rightarrow A$

Tasks:

1. Compute A^+ .

Ans - $A^+ = \{A, B\}$

2. Compute BC^+ .

Ans - $BC^+ = \{A, B, C, D, E\}$

3. Identify whether A functionally determines all attributes of R.

Ans - A does NOT determine all attributes of R.